Sre Cibi Varsshan K

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EDUCATION

Vellore Institute of Technology, Chennai *Expected 2027*
B.Tech Computer Science and Engineering (Artificial Intelligence and Machine Learning)
Relevant Coursework: Computer Graphics, Data Structures and Algorithms, Computer Vision.

SUMMARY

C++ developer focused on native and GPU code: FFmpeg C API video processing, OpenGL fragment and compute shaders, ray marching and ray tracing, and JNI native code on Android. Learning Vulkan.

PROJECTS

censface-cpp – Video Processing Tool *Oct 2026*
C++17, FFmpeg, OpenCV, CMake, Linux
github.com/xerfzy/censface-cpp

- Implemented a single-pass pipeline on the FFmpeg C API: libavformat demux, libavcodec send/receive decode and encode with end-of-stream draining, libswscale YUV-to-BGR conversion with explicit colour matrix and range, and interleaved MP4 muxing.
- Kept variable-frame-rate timing by encoding in the input stream's time base with per-frame timestamps and rescaling packets to the muxer's time base; stream-copied audio after checking MP4 codec support.
- Wrapped every FFmpeg object in RAII `unique_ptr` deleters with atomic temp-file output and SIGINT handling; verified identical frame timestamps, byte-identical audio and flat peak memory (486 vs 490 MB for 6 s vs 29 s of 720p HEVC).

ChaCha20 GPU Image Encryption *Mar – May 2026*
C++17, OpenGL 4.3, GLSL Compute, SSBO
github.com/xerfzy/chacha2020ImageEncryption$_u sing_c ompute_s hader$

- Implemented the ChaCha20 quarter-round and 20-round block function in a GLSL compute shader (local size 256); each invocation produces two 64-byte keystream blocks from its counter and XORs them into the SSBO in place.
- Packed decoded images into one 128-byte-aligned buffer per batch with per-image offsets, planning batches from stb_image header scans against a VRAM budget, then ran upload, dispatch, shader-storage barrier and mapped readback.

SDF Ray Marcher *Jan – Apr 2026*
C++, OpenGL, GLSL, GLFW, GLM
github.com/xerfzy/Ray$_m archer_u sing_f ragment_s hader_i n_o pengl$

- Built a real-time sphere-tracing renderer in a fragment shader over signed distance functions (sphere, box, plane) with domain repetition, union and central-difference normals, driven by a free-fly camera passed as uniforms.
- Added a two-pass pipeline that renders into a framebuffer object and applies a Sobel edge-detection pass; also wrote a compute-shader version that writes an RGBA32F image through `imageStore`.

CPU Ray Tracer *Nov – Dec 2025*
C++14, CMake
github.com/xerfzy/Ray$_t racer - learning - project-$

- Implemented a ray tracer following *Ray Tracing in One Weekend*: Lambertian, fuzzy metal and dielectric materials (Snell refraction, Schlick reflectance), jittered multi-sample anti-aliasing, recursive bounces and defocus blur from a thin lens camera model.

XrivenixAI – Native Inference on Android *Sep 2026*
C++, JNI, Android NDK, CMake, Kotlin
github.com/xerfzy/XrivenixAI

- Wrote the JNI bridge to llama.cpp (built from source with the NDK for arm64-v8a): model loading, batched decoding, streaming generation, grammar-constrained completion and KV-cache prefix reuse across turns.

eos-oneplus-nord *Oct 2026*

Bash, Linux, adb, fastboot

github.com/xerfzy/eos-oneplus-nord

- Wrote Linux tooling that SHA-256-verifies resumable downloads, checks device state over adb and fastboot, and audits a vendor installer against pinned hashes before allowing a guarded flash.

RESEARCH

Indian Patent Application IN202641060337 A1 *May 2026*

Published application, not granted

Title: "A System for Swarm of Drone for Crowed Behaviour Analysis"
Application No. 202641060337, filed 12 May 2026, published 22 May 2026.
Inventors: Sre Cibi Varsshan K, D Kavitha.
Applicant: Vellore Institute of Technology, Chennai.

TECHNICAL SKILLS

Languages: C++ (C++17), C, GLSL, HLSL, Bash, Python, Kotlin

Graphics and GPU: OpenGL 4.3, fragment and compute shaders, SSBO, framebuffer objects, image load/store, ray marching, ray tracing; Vulkan (learning: Vulkan-Hpp, swapchain, Slang/SPIR-V)

Libraries: FFmpeg (libavformat, libavcodec, libswscale), OpenCV, GLFW, GLM, Dear ImGui, OpenSSL, stb_image, llama.cpp

Tools: Linux, GCC, CMake, pkg-config, Git, Android NDK, Visual Studio